MATLAB Audio Sampler

User Guide and Feature Reference

Project Notes

- More information about certain features are available in the code. The original project documentation asks the reader to open `AudioSampler`; an extra GUI copy was included as a backup in case of breakage.
- The original instructions suggest using a YouTube-to-MP3 website to obtain MP3 audio files. Use audio files that you have permission to download and use.

Pads / Sample Slots

- **Pad (1–9):** Each pad is a playback button. When a sample is loaded into a slot, pressing its corresponding pad starts audio playback.
- **Load (1–9):** These buttons allow the user to load an audio sample of their choice (preferably `.wav` or `.mp3`). When a sample has been loaded, the selected Load button turns green to indicate that the slot contains audio. Press the corresponding Pad button to play the sample.

Tone Generator

- **Waveform:** The waveform dropdown selects the shape of the sound wave used to generate the tone. Different waveforms produce different sound characteristics, as shown below.

Waveform	Description	Sound Character
Sine	Smooth continuous wave	Pure tone
Square	Alternates between high and low values	Hollow / Electronic
Saw	Ramp-shaped waveform	Bright / Buzzy
AM	Amplitude-modulated sine wave	Pulsing / Tremolo sound

- **Frequency (Hz):** This field determines the pitch of the generated sound. For example, 440 Hz corresponds to the musical note A. In general, higher frequencies produce higher pitches.
- **Duration (s):** This field determines how long the generated sound lasts before playback stops.
- **Generate:** This button generates a tone using the chosen waveform, frequency, and duration settings.
- **SynthSelection:** This dropdown chooses which sampler pad (1–9) will receive the generated tone.
- **Save:** This button saves the generated tone to the pad selected in the SynthSelection dropdown.

Equalizer

The equalizer adjusts the strength of different frequency ranges in the audio signal. Each slider boosts or reduces a particular band of frequencies.

- −12 **dB:** Reduces the frequency band.
- **0 dB:** No change.
- +12 **dB:** Boosts the frequency band.

Frequency	Characteristics
63 Hz – Bass / Sub-bass	Deep bass, rumble, kick-drum energy
250 Hz – Low / Midrange	Warmth, fullness, lower instruments
1 kHz – Midrange	Vocals, instrument definition, tone balance
4 kHz – Upper Midrange / Presence	Sharpness, consonants, articulation
8 kHz – Treble / High Frequencies	Brightness, sparkle, hiss

Note: After changing the EQ settings, you must replay or reload the sample for the changes to take effect. The equalizer does *not* update the sound dynamically during playback.

Audio Processing

- **Slot:** This dropdown selects which audio pad (1–9) the processing controls apply to. The slot must contain loaded audio before processing can occur.
- **Start / End / Apply Chop:** The Start field defines the beginning of the audio segment to keep, measured in seconds from the start of the sample. The End field defines where the segment stops. Apply Chop trims the audio to the selected interval. For example, setting Start to 1.5 s and End to 4 s keeps the audio from 1.5 s to 4 s.
- **Echo / Delay Time / Feedback:** The Echo switch enables or disables the echo effect. Delay Time controls the time, in seconds, before an echo is heard: smaller values produce rapid echoes, while larger values create more widely spaced echoes. Feedback controls how much of the echo is fed back into the system; higher feedback creates multiple repeating echoes.
- **Reverb:** The Reverb switch controls the strength of the reverberation effect. Higher values increase the number and strength of reflections. Lower settings can provide a subtler effect.
- **Direction / Volume / Speed:** Direction changes the playback direction of the audio sample. Volume adjusts loudness (a setting of 2 can be quite loud), and Speed changes the playback rate.
- **Pause or Resume / Stop:** Pause or Resume temporarily pauses playback while preserving the current position. Stop ends playback completely and resets the pad color to gray.

Waveform and Synth Graph Displays

- **Waveform:** The Waveform graph displays the audio signal from the selected pad after processing, including any applied effects.
  - **X-axis:** Time (s), showing how the audio progresses over time.
  - **Y-axis:** Amplitude, showing the signal strength at each moment.
- **Synth:** The Synth graph displays the waveform of the generated tone. After pressing Generate, the graph shows the mathematical waveform created using the selected settings.
  - **X-axis:** Time (s), showing the duration of the generated tone.
  - **Y-axis:** Amplitude, showing the height of the waveform.